11.3 Areas of Polygons

Name _____ Period ____ Date _____

Geometry 11.3 (30-minute plan) · Student sheet

I can…

- find the area of a rhombus or a kite from its diagonals.
- find the central angle of a regular polygon.
- find the area of a regular polygon using its apothem and perimeter.

Vocabulary · fill in the missing word

Center of a regular polygon — the center of its _____ circle

Radius of a regular polygon — the radius of its _____ circle

Apothem of a regular polygon — the _____ from the center to any side of the polygon

Central angle of a regular polygon — an angle formed by two radii drawn to _____ vertices

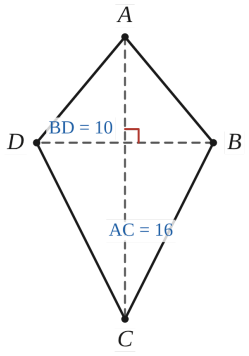
Part 1 · Notes and examples (fill in each note, then work the example)

Area of a rhombus or kite: $A = \frac{1}{2}$ _____, where d_1 and d_2 are the diagonals.

Example 1

a. Find the area of the kite.

b. A rhombus has diagonals of 6 and 9. Find its area.

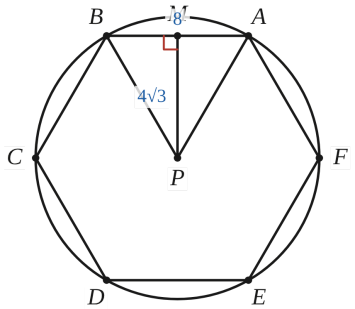


Central angle of a regular n-gon: _____

Area of a regular polygon: $A = \frac{1}{2} \cdot a \cdot P$, where a is the _____ and P is the _____.

Example 2

A regular hexagon has sides of 8 inches and an apothem of $4\sqrt{3}$ inches. Find its area. Round to the nearest hundredth.



Example 3

A stop sign is a regular octagon with sides of 10 inches and an apothem of about 12.07 inches. Find its area. Round to the nearest tenth.

Part 2 · Partner practice

Partner practice

A regular hexagon has sides of 6 and an apothem of $3\sqrt{3}$. Find its area. Round to the nearest hundredth.